

Progress Report: Creation and Evaluation of Bifunctional Organic Diodes for Light Emission and Detection

Thomas Cox, Ajay Pandey, Maciej Klein

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Abstract

This report covers the first semester of a two semester project that aims to recreate and expand upon the organic bifunctional devices that are capable of both emitting (OLED) and absorbing (OPD) light signals based on the pre-established design by Marcin Kielar [1] and characterise it in a laboratory setting investigating key characteristics such as I-V measurements, linearity tests, and irradiance. Using equipment at QUT's Central Analytical Research Facility (CARF), these devices were fabricated. Using them, data was measured within an isolated blackout Faraday cage to reduce external noise. This data allowed for the plotting and examining of what the strengths, such as flexibility, efficiency, and adaptability, as well as the limitations such as the degradation due to being organic. These results provide a strong start to the project and provide a clear path forward. During the next semester three key characteristics will be investigated: Reproducibility, preventing degradation, and exploring various materials to expand upon customisation.

Keywords:

1 Introduction

As part of a larger project in exploring electron spin manipulation, this project aims to address the practical engineering challenge of making organic bifunctional devices which operate in two modes, light emission (OLED) and light detection (OPD). These devices are suitable for things such as tactile sensing and wearable applications and come with the several key advantages such as being flexible, lightweight, and able to be fine-tuned for specific applications. Despite this, they suffer from performance variability and environmental degradation that limit reliability. This work aims to fabricate prototype devices based on the pre-established design by Marcin Kielar [1] and

characterise it in a laboratory settings investigating specifically OLED/OPD I-V measurements, linearity tests, and calibration of photo-diode current to irradiance to deliver processed data, analysis scripts, figures, and practical recommendations. Limitations to note are the results are from lab-scale devices which house only 5 pixels on a single 6.25cm^2 square surface. Because of this, array scaling and production issues are not addressed. Additionally, the sample size is limited which restricts statistical generality.

2 Background and Literature

2.1 Context and Relevance

Organic optoelectronic devices that combine light emission and photodetection into a single pixel are an enabling technology for flexible, lightweight sensing for use in areas such as tactile sensor and electronic skin. These devices can be fabricated with low-cost thin-film techniques and reconfigured to operate as emitters (OLED), and detectors (OPD), or energy harvesters, making them attractive for wearable and soft-robotic applications [1, 2]. The present project focuses on reproducing Kielar-style bifunctional device architecture and producing electrical-optical characterisation to support device optimisation.

2.2 Key device principles

2.2.1 OLED operation

An OLED produces light when electrons and holes injected from opposite electrodes recombine within an emissive organic layer to form excitons. Device performance is governed by charge injections and transport, exciton formation and quenching, and optical outcoupling. Typical electrical signatures include an I-V curve similar to that of a diode but with a distinct turn-on voltage and exponential increase in current above that threshold [3]

2.2.2 OPD Operation

An OPD converts incident photons into an electrical signal. This occurs as photon absorption creates excitons that dissociate into free charges at donor-acceptor interface or at heterojunctions. These collected charges produce a photocurrent. Important detector behaviours include dark current, responsivity, detectivity, and linearity of photocurrent with irradiance. [4, 5]

2.3 Typical Performance Metrics

Typical performance metrics reported in the literature include the OLED turn-on voltage (V_{on}) and I-V characteristics, which are often plotted on a logarithmic current axis. Electroluminescent output such as luminance (cd/m^2), spectral power distribution, external quantum efficiency (EQE). OPD figures also typically report values such as dark current, short-circuit photocurrent (I_{SC}), responsivity (A/W), noise equivalent power (NEP), specific detectivity (D^*), and linear dynamic range. Device stability is typically characterised by lifetime metrics and by changes in I-V

behaviour or responsivity following ageing or environmental exposure [1, 5]

2.4 Gaps addressed by this project

Although prior work reports device structures and example measurements, several practical gaps remain for system integration and engineering assessment:

- Quantitative, calibrated mapping between OLED electrical properties (current and voltage) and emitted irradiance (W/cm^2 for the specific Kielar derived device).
- Practical characterisation of OPD linearity and usable dynamic range under controller irradiance levels relevant to tactile-sensing use cases.
- implementation details for measurement calibration and uncertainty (photodiode responsivity, area, alignment) which are often omitted or under-documented in published examples

3 Project Objectives and Deliverables

3.1 Semester Objectives

The objectives for this semester are:

- Replicate the bifunctional device structure established by the reference paper, ensuring that the process matches the design as closely as possible.
- Developing proficiency with the data acquisition processes which would allow me to reliably measure and test the devices.
- Examine how device degradation affects performance over time which is crucial for assessing the practical viability and long-term reliability of these organic systems

3.2 Interim Deliverable Achieved

The following deliverables have been completed in this first semester:

- Fabrication: Produced a batch (4) of multi-pixel devices (5 pixels on a 6.25cm^2 substrate). **Completed:** 26-9-2025
- Measurement Proficiency: Implemented automated measurements for voltage sweeps and data logging. **Completed:** 20-10-2025

- Degradation Effect Quantisation: observed and plotted a rudimentary display of the effects of degradation. **Completed: 22-10-2025**

3.3 Planned Future Milestones

- Expand sample size by fabricating additional devices and repeat measurements to assess reproducibility.
- Further investigate the degradation process to determine the harshest causes and attempt to resolve this to provide a longer lifetime.
- Determine other combinations of materials to gain true control and customisation of the device.

4 Methods

4.1 Device Fabrication

Devices were fabricated at QUT's Central Analytical Research Facility (CARF) on ITO-coated glass substrates based on the device shown in fig. 1. Substrates were cleaned with UV-ozone, which allowed for the following spin-coating (at 5000rpm for 60seconds) to evenly distribute the PEDOT:PSS liquid across the surface. Once coated it was then baked on a hotplate (120C for 10 minutes) to dry and solidify the PEDOT:PSS. The remaining organic and metal layers were deposited by thermal evaporation in a vacuum chamber with temperatures ranging from 200 $\rightarrow$ 400C for several hours depending on the material being evaporated. Two of the evaporated layers were patterned through a shadow mask to define directed electrode lines and pixels which can be seen on the finished product in fig. 2. Following fabrication the devices were stored in a cool dimly lit chamber filled with nitrogen to prevent and oxidation and degradation prior to testing.

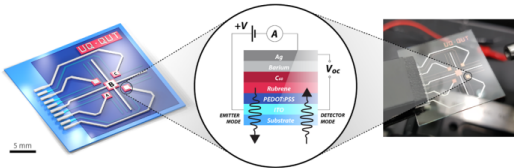


Figure 1: Device structure obtained from the reference paper [1]

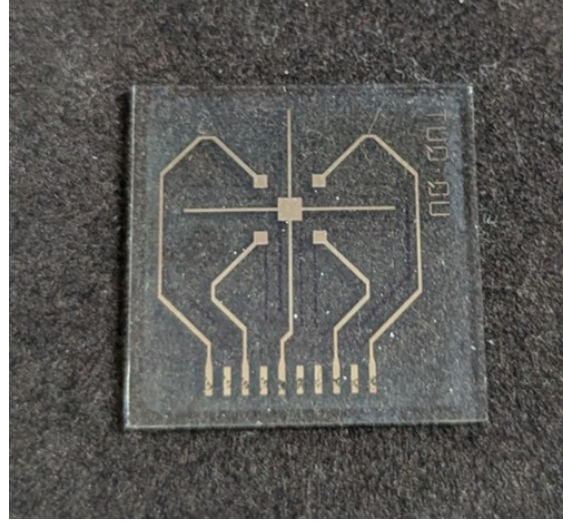


Figure 2: Fabricated device

4.2 Device Characterisation

To characterise each device, they had to be removed from the nitrogen containment chamber as the setup required (fig. 3) was in a separate building (S905B). this setup used a laptop with LabVIEW directly connected to a digital multimeter which was able to read the electrical data in both current and voltage driven modes with the device placed in a blackout Faraday cage to reduce any external noise.

The measurement that were taken consisted of the I-V curves in both OLED and OPD operating modes, which allowed for the performance of the device as an emitter and detector to be assessed. Linearity tests were then completed by varying the incident irradiance and measuring the corresponding electrical response to evaluate how the drive output scales with light. The exact setup and parameters can be seen in table 1

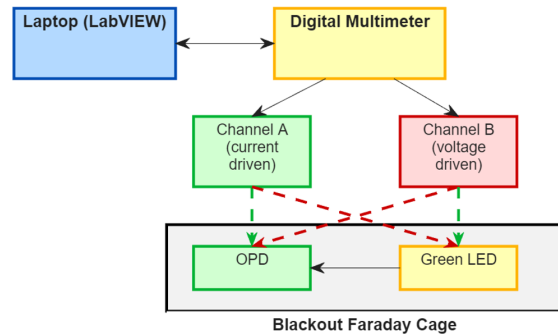


Figure 3: Characterisation Setup

Table 1: Measurement protocols and sweep parameters

Measurement	Drive mode	Distance to detector (mm)	Sweep / Protocol (clear statement)
OPD V-I (dark / illuminated)	Voltage-driven (SMU applies bias; measure current)	50 mm	Voltage sweep from -1.00 V to 1.00 V in 0.01 V steps; measure dark I-V and illuminated I-V (illuminated run at 10 W/m^2); use dark subtraction to obtain photocurrent.
Linearity — photocurrent measurements	Light-driven at fixed bias (bias = 0 V ; photocurrent measured)	50 mm	Irradiance sweep across decades (approx. 1×10^{-5} to $2.5 \times 10^2\text{ W/m}^2$); at each irradiance record 5 measurements per point, 1 cycle; wait 3 s before start; compute mean and extended standard deviation (applied to first 5 points).
Linearity — open-circuit voltage (V_{oc})	Light-driven open-circuit measurement (OPD open-circuit; LED provides illumination)	50 mm	Measure V_{oc} vs irradiance: 5 measurements per point, 1 cycle; wait 5 s before start; extended standard deviation applied to first 10 points. OPD measured using Agilent multimeter (USB connection).
OLED I-V and simultaneous optical mapping	Voltage-driven (SMU applies voltage; measure OLED current)	50 mm (photodiode facing OLED)	Voltage-driven sweep from 0.0 V to 5.0 V in 0.1 V steps; device compliance/current limit set to 15 mA ; record OLED current and photodiode signal simultaneously to build electrical-to-optical calibration ($I_{OLED} \rightarrow \text{irradiance}$).

5 Results

The following plots correspond to the previously mentioned measurements:

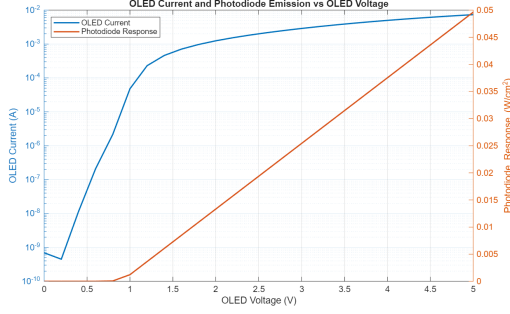


Figure 4: Electrical-to-optical mapping: photodiode response vs. OLED drive

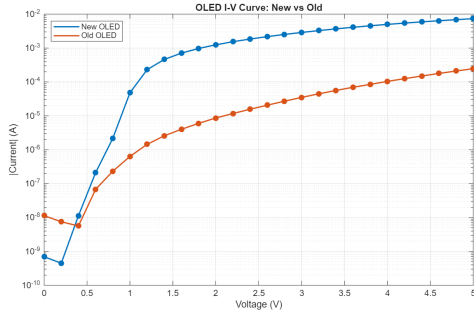


Figure 5: OLED I-V curve (new and aged devices)

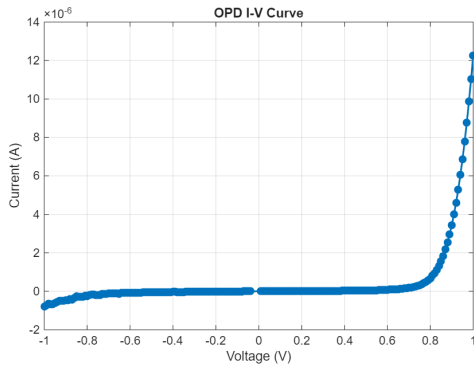


Figure 6: OPD dark I-V characteristic

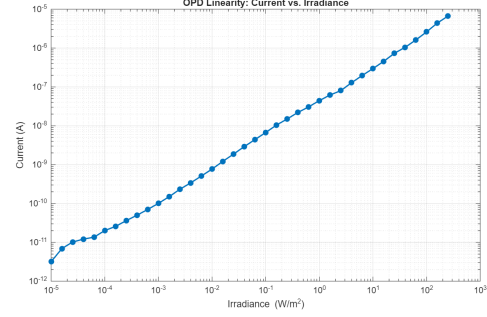


Figure 7: OPD photocurrent vs. irradiance (linearity test)

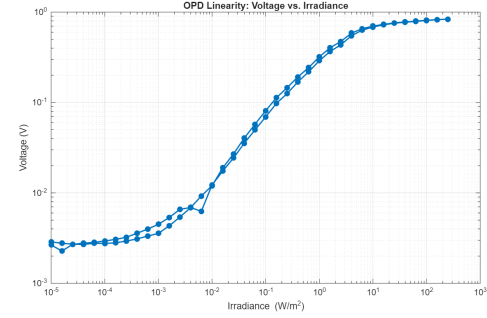


Figure 8: OPD voltage response / V_{oc} vs. irradiance

6 discussion

6.1 OLED Results

fig. 4 demonstrates the relationship between electrical and optical performance of the device in OLED mode. On the left axis it can be seen how the OLED current increases rapidly once past the turn-on voltage threshold, demonstrating typical behaviour for diodes. On the right axis, the photodiode response tracks the amount of light emitted by the device, which, once the device is active, increases near linearly with the applied voltage. This clear correlation between current and emission highlights the devices efficient electrical-optical conversion allowing how changes in voltage translate to light output to be directly assessed.

6.2 Device Degradation

To start to understand the effect degradation has on these devices, an older sample (unknown how old) was compared to this newer device (fig. 5). As shown, there are clear differences for these two devices. For instance, the new device carries much higher forward current across the whole bias range. Additionally, the new device turns on at a lower range (0.8V compared

to 1.2V using a practical threshold of $1 \times 10^{-6} A$). The steeper turn-on slope of the new device indicates improved charge injection/transport, whereas the old device's reduced current higher effective Von are consistent with degradation. Practically, the new device will produce higher light output at a given drive but at the cost of greater power draw, while the old devices' performance loss points to the need for encapsulation. These observations are a key motivation for future testing on this degradation.

6.3 OPD Results

fig. 6 shows the I-V curve for the OPD. As expected for a diode, the current remains very low at negative voltages, indicating low leakage or dark current. Once a positive voltage is applied the current increases exponentially, which is characteristic of good diode behaviour. This exponential response is important as it confirms that the device operates efficiently as a photodetector, with strong rectification and low background noise. When looking at the negative bias, the dark current is especially critical for resistive detection as it helps distinguish true signals from noise. Overall, these results demonstrate that the OPD is functioning as intended providing the necessary electrical characteristics for practical light sensing applications.

6.4 Linearity Results

fig. 7 and fig. 8 display the current and voltage output of the OPD as a function of irradiance. In fig. 7, the current increases near linearly, with minor fluctuations due to things noise, while on a logarithmic scale. This indicates that the OPD maintains a proportional response to light intensity over several orders of magnitude. This strong response is ideal for things such as quantitative sensing. fig. 8 displays that the voltage also increases with irradiance. However, this relationship is only linear within an intermediate range of $10^{-3} \rightarrow 10^1 W/m^2$. At very low irradiance, the voltage response is limited by noise and device sensitivity while at high irradiance, the voltage begins to saturate indicating its reached its peak voltage for any given irradiance higher than $10 W/m^2$. These results highlight the importance of identifying the usable linear range for both current and voltage as accurate and reliable sensing depend on operating within this range.

7 Risk Management, Ethics and Sustainability

7.1 Risk and safety

All experimental activities are conducted under the laboratory safety framework of QUT and CARF. What they are and how they were handled are displayed in Appendix C and contain things such as harmful chemicals, high temperatures, electrical shock, etc.

7.2 Ethics

This research adheres to the principles of responsible and reproducible research. Data integrity is enforced through versioned data storage, retention of raw measurement files, and documented processing scripts so results can be independently reproduced and audited.

7.3 Sustainability

This project seeks to minimise environmental impacts associated with the materials and processes used. Key impacts include solvent use and chemical waste from solution processing as well as energy consumption by the machines used within both CARF and the S905B laboratory.

8 Resources Used

The resources used for this project can be put into several categories such as location, major equipment (fabrication), major equipment (characterisation), supporting instruments, etc. The primary equipment used consisted of an ozone cleaner, heat plate, nitrogen filled glovebox, vacuum chamber, and the several materials used for the fabrication. A full list can be seen in Appendix C.

9 Conclusions and Next Steps

Overall this project indicates steady progress with a clear path forward and has successfully assisted in the fabrication and broad characterisation of a bifunctional device that operates as both an OLED and OPD. Through a series of experiments, it was possible to analyse the device performance and investigate linearity and sensitivity which gave a clear picture of how the device behaves and what its strengths and limitations are.

Following a brief break, this project will continue in semester 1 2026 with the following goals (in order of

priority):

1. Expand sample size and verify reproducibility.
2. Investigate degradation and controlled lifetime through the means of encapsulation.
3. Explore the variability of these devices through the use of different materials.

9.1 Deliverables for the next report

By the next milestone the project will deliver:

- An expanded datasets with repeatability statistics
- Results from at least one encapsulation/ageing experiment
- Information on materials and their effects on organic diodes.

10 References

References

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A Appendix A: Updated Timeline and Deliverables

- Original milestone table (copy from proposal).
- Completed items with dates.
- Changes to timeline and justification.

B Appendix B: Raw Data Files and Processing Notes

- List of raw data files (filenames, brief description).
- Data processing scripts used (LabVIEW, MATLAB, Python); include short pseudocode or attach code in supplementary files.
- Notes on unit conversion and calibration factors (responsivity, area).

C Appendix C: Safety Data Sheets & Resources Used

Table 2: Key laboratory hazards and how they were handled

Hazard	How it was handled / Controls implemented
Chemical solvents and process chemicals (acetone, isopropanol, PEDOT:PSS solutions, organic semiconductors)	Work performed in a fume hood; SDS sheets available at point of use; nitrile gloves, lab coat and safety glasses worn; solvents dispensed from labelled containers and waste collected in segregated, labelled chemical waste bins for authorised disposal.
Organic powder handling and sensitive materials (rubrene, C ₆₀)	Minimised exposure to air/moisture by handling in glovebox or quickly within a fume hood; use of clean tools/paper; PPE (gloves, mask if dust risk); material receipts and purities logged; small batch quantities to reduce waste.
UV-ozone cleaning and UV exposure	UV-ozone hood used with shield closed; operators trained in equipment SOP; avoid direct line-of-sight exposure; use of UV-rated eye protection where required.
Thermal evaporation / vacuum systems	Only trained personnel operate evaporator; chamber interlocks and emergency shutoff in place; base-pressure and leak checks performed before runs; quartz crystal monitoring used for thickness control; hot surfaces identified and gloves used when unloading.
Broken glass / substrate handling	Use of cut-resistant gloves when handling substrates; protect work surfaces; broken glass cleaned up with brush & pan and disposed to sharps/glass disposal; visual inspection of substrates prior to use.
Electrical hazards from SMUs and test rigs (biasing, high current)	Instrument compliance/current limits and grounding enforced; insulated probe fixtures used; measurements performed inside light-tight / Faraday enclosures; never work on live circuits without PPE; supervisory approval for high-power tests.
Bright LEDs / optical sources (eye hazard)	Measurements performed inside blackout enclosures when possible; beam paths minimised; eye protection used if exposure risk exists; avoid direct viewing of LEDs.
Compressed gases / nitrogen (storage, glovebox)	Glovebox/N ₂ cabinet operated per SOP; room ventilation and oxygen monitors where required; cylinders secured and handled with proper regulators; emergency procedures for asphyxiation risk documented.
Flammability and fire risk	Flammable solvents stored in flammable cabinets; no open flames near solvent handling; fire extinguishers and spill kits accessible; staff trained in fire response and evacuation routes.
Waste and environmental hazards	Chemical wastes segregated and labelled; metal/organic scrap collected for appropriate disposal or recycling; minimised solvent volumes and batched runs to reduce energy/waste.
Emergent incidents (equipment fault, chemical spill, injury)	Work paused and area isolated; supervisor and CARF safety contacted immediately; incident logged through CARF/university reporting system; corrective actions implemented before resuming work.

Table 3: Resources used for device fabrication, characterisation and analysis

Resource category	Item (example / model where known)	Purpose / notes
Facilities	QUT Central Analytical Research Facility (CARF); laboratory bench space; black-out/Faraday enclosure	Primary locations for substrate preparation, thin-film deposition and electrical/optical testing. Access under CARF booking and supervision.
Major equipment (fabrication)	Spin coater; UV-ozone cleaner; thermal evaporator (high vacuum); hotplate; shadow masks	Used for substrate cleaning, PEDOT:PSS coating, organic/metal deposition and pixel definition. Operated by trained personnel per SOPs.
Major equipment (characterisation)	Source-Measure Unit (SMU, e.g., Keithley SourceMeter); Agilent 3450A multimeter; calibrated silicon photodiode; spectrometer / photometer (if available); neutral density filters	Electrical I-V sweeps, photocurrent and Voc measurements, optical power/spectral measurements, and LED/irradiance control. Record instrument calibration status in Appendix.
Supporting instruments	Vacuum pump and gauges; quartz crystal microbalance (thickness monitor); glovebox / N2 cabinet; fume hood	Used for vacuum evaporation control, thickness monitoring, oxygen/moisture sensitive handling, and safe solvent handling.
Software & data tools	LabVIEW / MATLAB / Python scripts; data logging formats (CSV, TXT);	Automated measurement control, data capture, processing scripts, and reproducible analysis. Maintain raw data and code repository.
Consumables & materials	ITO-coated glass substrates; PEDOT:PSS; organic semiconductors (e.g., rubrene, C ₆₀ or equivalents); metal sources (Ag, Al); solvents (acetone, IPA); desiccants; shadow masks	Sourced from lab stock or suppliers. Track lot numbers, concentrations and storage conditions for reproducibility. Minimise waste and batch runs where possible.
Personnel & supervision	Student researcher (Thomas Cox); supervisors (A/Prof Ajay Pandey, Dr Maciej Klein); CARF technician support	Roles: device fabrication, measurement, data analysis, and oversight. CARF technicians assist with equipment operation and maintenance.
Data storage & backup	Lab server / university OneDrive; local backup drive; raw data folders (Appendix B listing)	Store raw and processed data with clear filenames, timestamps and versioned analysis scripts for reproducibility and audit.